

Closed-Loop Long-Tail Synthesis for Driving Vision-Language Models

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Abstract—The deployment-safety frontier for autonomous vehicles is the long tail. Vision-language models (VLMs) for driving — Waymo’s EMMA, DriveVLM, GPT-Driver — are bottlenecked by exposure to the rare scenes that demand reasoning. Real long-tail clips constitute well under 0.1% of any open driving corpus. This proposal describes a full research program to address that bottleneck with a closed-loop mining-generation-verification pipeline. *Stage A — Mining* embeds clips with DINOv3 and V-JEPA-2.1, runs a 5-model BEV-detector ensemble (UniAD, VAD, BEVFormer, PARA-Drive, StreamPETR), computes the Westhofen criticality suite over 16 trajectory metrics, tags scenes with Cosmos-Reason2-8B, and aggregates into a calibrated multi-axis high-signal score. *Stage B — Layout extraction* converts each high-signal seed into a typed `SceneLayout` via dataset-specific extractors (Waymo, nuScenes, nuPlan, Argoverse 2, CODA, comma2k19). *Stage C — Generation* applies a 5-axis perturbation taxonomy (environmental, agent substitution, agent multiplication, trajectory, layout) and renders 20 variants per seed through Cosmos-Transfer-2.5-2B (single front view), Cosmos-Single2MultiView (8-camera surround), Cosmos-Transfer-LidarGen (matching synthetic LiDAR), and a UniAD auto-labeler. *Stage D — Verification* applies four gates (realism, Cosmos-Reason2-8B semantic match, HS preservation, customer-model failure) and ships verified bundles in nuScenes, Waymo, and OpenSCENARIO formats. The flagship validation experiment trains BEVFusion from scratch on real-only versus real-plus-synthetic recipes across four data designs, with pass criteria registered in advance. At full scale on the WOD-E2E corpus (4,021 hand-curated long-tail seeds $\times$ 20 variants $\times$ $\sim$ 50% gate survival), the program produces $\sim$ 40K verified high-signal synthetic clips with full provenance. We argue this is the highest-leverage open-source contribution available in synthetic AV data in 2026, and that the closed loop is the moat: targeting plus filtering, neither of which the existing literature provides.

Index Terms—autonomous driving, vision-language models, long-tail mining, controllable video generation, synthetic LiDAR, multi-view consistency, closed-loop learning, Cosmos-Transfer, UniAD, BEVFusion.

I. INTRODUCTION

A robotaxi can drive 99.9% of urban miles cleanly and still fail on the 0.1% that ships it to regulators [1]. The events that determine deployment safety — occluded VRUs, unprotected turns under glare, dust storms, school bus passing children, India-style unstructured traffic, flooded roads — are statistically vanishing in any open corpus. Vision-language models for driving (Waymo’s EMMA [2], DriveVLM [3], GPT-Driver [4], the broader VLA-for-AD line [26]) are particularly exposed: their reasoning capacity is gated by exposure to exactly the scenes the data does not contain.

Two recent developments make the long-tail gap addressable for the first time. First, NVIDIA’s Cosmos-Drive-Dreams pipeline [8] and the Cosmos-Transfer-2.5-2B follow-up [9] demonstrate that pretrained controllable video generation can produce photoreal driving clips conditioned on HD map polylines, 3D cuboids, and free-text prompts — and that these generated clips improve downstream perception by up to 60% on detection metrics. Companion checkpoints handle multi-view extension (Cosmos-Single2MultiView) and matching synthetic LiDAR (Cosmos-Transfer-LidarGen). Second, foundation-model embedding stacks (DINOv3 [11], V-JEPA-2.1 [12]) plus AV-aware reasoning VLMs (Cosmos-Reason2-8B [10]) make it cheap to score scenes along multiple axes — novelty, difficulty, safety, semantic match — that previously required bespoke per-axis classifiers.

What is still missing is a *closed loop* that connects the two ends: a system that uses the scoring stack to mine seeds, the generation stack to manufacture variants, and the same scoring stack to reject generations that lost the property of their seed. NVIDIA’s pipeline stops at unconditional bulk generation. Academic mining baselines — VLMine [16] and the comprehensive “Mining the Long Tail” line [17] — stop at selecting from existing data. Neither targets the union; neither closes the loop with a downstream consumer model.

This proposal describes a full research program to build, validate, and release that closed loop. The technical contribution is the integration: a multi-axis mining-generation-verification pipeline that keeps long-tail signal across the round trip, instrumented to feed a downstream VLM-driving model. The empirical contribution is a controlled validation that quantifies, for the first time, how much of the gain comes from each component — targeting, generation, filtering — at scale. The pragmatic contribution is a $\sim$ 40K-clip public synthetic dataset with provenance, multi-view + LiDAR + auto-labels, in formats compatible with existing AV training pipelines.

II. BACKGROUND AND RELATED WORK

A. Long-Tail Mining for Autonomous Driving

Mining hard scenes from large unlabeled driving corpora has been studied along three orthogonal axes [17]. *Visual novelty*: distance in a learned embedding space — e.g., DINOv2/v3 nearest-neighbour or Mahalanobis — flags scenes that look unlike the training distribution. *Detector difficulty*: ensemble disagreement among SOTA 3D detectors (UniAD

[13], VAD [14], BEVFormer [15]) reveals scenes the planner cannot confidently understand. *Safety criticality*: the Westhofen criticality catalogue [27] formalizes 16 trajectory-based metrics (TTC, PET, TTB, TIT, TET, WTTC, MPrISM, etc.) that identify dangerously close encounters. Recent work — VLMine [16] and the comprehensive “Mining the Long Tail” comparative study [17] — combines two or more of these axes and reports 4–10% relative mAP gains on rare classes when the mined subset is used to fine-tune a perception model.

This program extends the literature in two ways. First, it adopts *five* axes — adding a rarity prior over a Cosmos-Reason2-tagged 200-leaf scenario taxonomy [10], and an information-gain proxy via Bayesian coreset / EPIG-style approximations — so that the scorer captures *why* a scene is long-tail, not just *whether* it is. Second, it grounds the scoring in a calibrated aggregate trained against realized downstream Δ mAP from controlled subset experiments, turning the scorer into a closed-loop optimizer rather than an open-loop selector.

B. Controllable Generation of Driving Scenes

Until 2024, controllable driving video generation required training a model from scratch — MagicDrive [18], Vista [19], DriveDreamer [20]. NVIDIA’s Cosmos-Drive-Dreams pipeline [8] changed this by releasing a pretrained AV-post-trained controllable generator, an 81,802-clip synthetic dataset (CC-BY-4.0), and the toolkit to render HDMap, LiDAR-depth, and 3D-cuboid conditioning videos. The Cosmos-Transfer-2.5-2B follow-up [9] is $3.5\times$ smaller than the 7B, runs at 5–10 minutes per 121-frame 720×1280 clip on a single H100, and outperforms the 7B on AV downstream metrics by up to 60%. Companion checkpoints — Cosmos-Single2MultiView and Cosmos-Transfer-LidarGen — extend the pipeline to 8-camera surround consistency and matching synthetic LiDAR sweeps. Together they form the cheapest credible AV-scene synthesizer available in 2026.

The proposed program adopts the full Cosmos stack as primary, with Vista and MagicDrive-V2 retained as ensemble alternatives for diversity (run in parallel on a fraction of seeds to verify that gate-survival rates and downstream gains generalize across generators). Crucially, no generation model is trained from scratch in this program — every generator is pretrained and open-weight.

C. Vision-Language Models for Self-Driving

The current frontier is not VLM-as-perception but VLM-as-reasoner. Waymo’s EMMA [2], DriveVLM [3], GPT-Driver [4], and the recent VLA-for-AD survey [26] all converge on a single architecture: a multimodal foundation model takes images plus a prompt, and emits either trajectories or driving instructions. Their measured failure mode is consistent: they reason fluently about common scenes and fail on long-tail ones. Adjacent work extends this line to off-road autonomy with vision foundation models (Velociraptor [5]) and to zero-shot 3D grounding via LLMs (SORT3D [6]); the TartanGround dataset [7] provides an off-road analog of nuScenes for evaluating these methods.

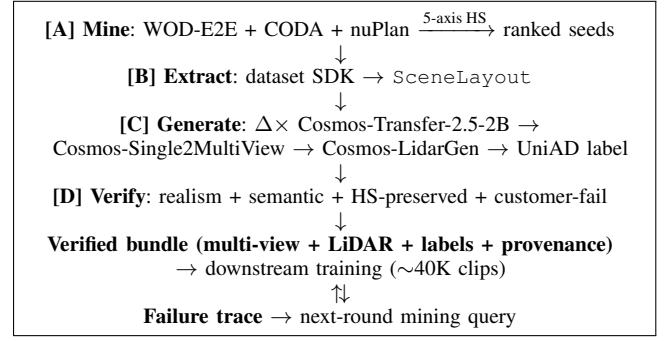


Fig. 1. The proposed four-stage closed loop, full-scale configuration: 5-axis high-signal scorer at Stage A, multi-view + synthetic LiDAR generation at Stage C, and four filter gates at Stage D.

D. The Gap and the Proposed Contribution

The gap is in the closure. Real-data mining (VLMine, Mining-the-Long-Tail) selects from a fixed pool. Controllable generators (Cosmos-Drive-Dreams, MagicDrive-V2, Vista) produce arbitrary clips without targeting specific failure modes. VLM driving research consumes whatever data is available without a feedback signal that says “these are the scenes I cannot handle”. The proposed program plugs into all three: a long-tail *generation* stage that augments real-data mining with controlled synthesis, gated by the same scoring stack used for mining so that synthetic outputs retain their long-tail property, and instrumented to use the downstream VLM’s residual failures as queries for the next round of mining.

III. SYSTEM ARCHITECTURE: FOUR STAGES

The proposed system (Fig. 1) is a four-stage pipeline that closes between a VLM driver under test and a long-tail data factory:

$$\underbrace{\text{Mine}}_A \rightarrow \underbrace{\text{Extract}}_B \rightarrow \underbrace{\text{Generate}}_C \rightarrow \underbrace{\text{Verify}}_D \rightarrow \text{train} \rightarrow \text{Mine again} \quad (1)$$

The novelty is not any individual stage — each piece adapts published work. The novelty is the closure: the same scoring stack mines seeds and gates generations, and the downstream VLM’s failures feed back as new mining queries, so the pipeline becomes a model-aware long-tail manufacturer rather than a generic data dump.

IV. STAGE A: MULTI-AXIS HIGH-SIGNAL SCORING

A. The Five Axes

For a scene s we compute a calibrated 5-tuple

$$\mathbf{HS}(s) = (\text{nov}(s), \text{diff}(s), \text{safe}(s), \text{rare}^{-1}(s), \text{IG}(s)) \quad (2)$$

plus a learned aggregate $\sigma(s) \in [0, 1]$.

Novelty. A 50/50 mix of (i) k-NN distance from s ’s DI-NOv3 [11] embedding to a real reference distribution (5K-frame nuScenes or Waymo train sample) and (ii) Mahalanobis distance from the reference centroid. V-JEPA-2.1 [12]

embeddings are computed in parallel and combined via an unweighted average to add temporal sensitivity. Both raw distances are robustly min-max scaled.

Difficulty. The disagreement of a 5-model BEV detector ensemble — UniAD [13], VAD [14], BEVFormer [15], PARA-Drive, StreamPETR — sourced from Bench2DriveZoo [24] and external repositories. Disagreement is measured as the variance of per-model 3D mAP on pseudo-labels and the variance of predicted ego trajectories. Per-model planning loss serves as a secondary signal.

Safety criticality. The Westhofen catalogue [27] of 16 trajectory metrics — TTC, PET, TTB, TTR, TIT, TET, WTTC, MPriSM, and others — computed on actor trajectories extracted from the detector ensemble. Aggregation uses a max over normalized per-metric scores so that any single near-miss criticality dominates.

Rarity. A frequency-inverted prior over a 200-leaf AV scenario taxonomy. Cosmos-Reason2-8B [10] tags each scene zero-shot with up to 5 leaf labels; the rarity score is the negative log-frequency of the scene’s tag set in a Pareto-fitted reference distribution.

Information gain. An EPIG-style [28] approximation of expected reduction in posterior uncertainty of a baseline detector if s is added to the training set, computed via a fixed-budget MC estimate on a calibration subset.

B. Aggregation and Calibration

The five raw axes are calibrated independently via isotonic regression against a small held-out set of clips with hand-rated long-tail labels. The aggregate $\sigma(s)$ is a learned ranker (XGBoost) trained on per-customer realized Δ mAP from controlled subset experiments — i.e., training a baseline detector on the top-K subset selected by a candidate σ and measuring how much that helps on a held-out long-tail eval. This is the closed-loop step that distinguishes the proposed scorer from open-loop selection methods.

C. Throughput and Cost

On a single H100 the full Stage A ensemble runs at ~ 10 K clips/hour. A full pass over the Tier-0 corpus (~ 1.4 M keyframes — WOD perception + WOD-E2E + CODA + nuPlan + comma2k19) takes ~ 140 H100-hours; embeddings and scores are cached forever, so this is a one-time cost.

V. STAGE B: LAYOUT EXTRACTION

Each high-signal seed is converted into a typed `SceneLayout` consumed by Stage C. The schema is dataset-agnostic and includes: scene identity and provenance, ego trajectory at 10 Hz, camera-rig spec, actor list with per-frame 3D-cuboid trajectories, HD map polylines (ASAM OpenDRIVE-compatible), environment metadata (weather, time of day, country, road type, illumination), the union of native dataset tags and Cosmos-Reason2-tagged scenario tags, an optional trigger event with timing, and the seed’s full HS profile. Per-dataset extractors are implemented for: Waymo Perception + WOD-E2E (Parquet v2.0.1 SDK),

nuScenes (nusenes-devkit), nuPlan (nuplan-devkit + scenario tags), Argoverse 2 (av2 SDK), CODA (COCO-format JSON + UniAD pseudo-3D-boxes), and comma2k19 (UniAD pseudo-labels + OpenStreetMap reverse-geocode for HD map). Throughput is ~ 5 sec/clip CPU-bound for parsing plus ~ 2 sec/clip GPU for UniAD pseudo-labels on unlabeled sources.

VI. STAGE C: PERTURBATION AND GENERATION

A. Five-Axis Perturbation Taxonomy

A perturbation Δ is a structured edit applied to a `SceneLayout`. Five axes are defined:

Environmental. Weather {clear, light/heavy rain, snow light/heavy, fog light/dense, dust storm, smoke wildfire, hail}; time of day {dawn, day sunny, day overcast, dusk, night lit, night unlit}; illumination special cases (direct sun glare, headlight glare, tunnel entry/interior, snow/water reflection); geographic style {US suburban, EU urban, IN unstructured, CN megacity, JP urban, ME dust, AU outback}.

Agent substitution. Swap an actor’s class while preserving its track: `vehicle.car` $\rightarrow$ `vehicle.truck.delivery`, `vru.pedestrian.adult` $\rightarrow$ `vru.pedestrian.child`, `vru.pedestrian` $\rightarrow$ `vru.cyclist`, etc.

Agent multiplication. Add or remove actors at BEV positions sampled from a learned scene-class-conditional prior (e.g., “more pedestrians on right sidewalk in school-bus scenes”).

Trajectory. Scale ego speed ($\times 0.5$ to $\times 2.0$, capped at 200 km/h peak), shift trigger event timing (± 2 s), vary actor velocities, lateral offset of the trigger from ego path.

Layout. Swap road geometry from a template library (4-way intersection, roundabout, highway entry, narrow 1-lane, construction zone with cone layout) while preserving scenario semantics; render via Cosmos’s HDMap conditioning.

B. Variant Sampling

Per seed, 20 variants are sampled from the Cartesian product of axis $\times$ operator with stratified coverage (at least one variant per axis). A validation pass rejects deltas that violate physics (agents through walls, ego speed exceeding the cap, agent dimensions outside class bounds) before generation. Per-customer overrides allow specific perturbation policies (e.g., “trucking customer wants only weather and ego-speed deltas, no layout swaps”).

C. Generation Pipeline

For each (layout, Δ_k) pair, the pipeline runs: (i) render HDMap, LiDAR-depth, and 3D-cuboid conditioning videos from the perturbed layout (CPU + GPU, ~ 10 sec/clip); (ii) build a deterministic text prompt from environment + scenario tags + delta description; (iii) generate single-front-view 121-frame 720×1280 video with Cosmos-Transfer-2.5-2B [9], ~ 5 –10 min on a single H100; (iv) extend to 8-camera surround with Cosmos-Single2MultiView, ~ 15 –20 min; (v) generate matching synthetic LiDAR with Cosmos-Transfer-LidarGen,

~10 min; (vi) auto-label the multi-view + LiDAR with the UniAD pipeline to produce 3D boxes, tracks, occupancy, and planning candidates that downstream models consume as training labels.

End-to-end per-variant runtime on 1×H100 with the 2.5-2B primary path: ~45–60 minutes, dominated by Single2MultiView. With 8×H100s running in parallel and Cosmos-LidarGen overlapped with the multi-view pass, throughput is ~10 verified clips per H100-hour at the assumed 50% gate-survival rate.

VII. STAGE D: FOUR-GATE VERIFICATION

Generated variants pass four gates:

Gate A — Realism. Three sub-checks: (i) physics sanity (UniAD’s auto-labels — do detected actors clip walls or have impossible velocities?); (ii) cross-sensor consistency (project 3D boxes from camera into the synthetic LiDAR frame, $\text{IoU} \geq 75\%$); (iii) distribution match (DINOv3 FID-style score against a real long-tail reference, aggressive-outlier rejection).

Gate B — Semantic match. Cosmos-Reason2-8B [10] verifies the generated multi-view depicts the requested scenario by emitting a numeric match score against the seed scenario tags plus the delta’s added tags. Threshold $\tau_B = 0.85$. Sister VLMs — Qwen2.5-VL-7B, InternVL — provide an ensemble vote when agreement is uncertain.

Gate C — High-signal preserved. Re-score the generated bundle on all five axes; reject if novelty ≥ 0.1 below seed novelty (drifted to common), if difficulty < 0.6 (SOTA detectors handle the variant), or if safety has fallen outside the seed’s target band (trigger event lost in generation).

Gate D — Customer-model failure. When a customer model is available via on-prem inference container, reject generations the customer’s model already handles correctly. Customer-specific gates make the pipeline maximally useful per-customer.

Compounded survival from Cosmos-Drive-Dreams paper estimates: ~50% of generated variants ship as verified high-signal. So 20 variants per seed yields ~10 verified — and 4,021 WOD-E2E seeds yield ~40K verified clips at the headline number.

VIII. OUTPUT BUNDLE AND ADAPTERS

Each accepted variant ships as a directory tree containing the 8-camera mp4s, synthetic LiDAR sweeps, 3D-box and trajectory auto-labels, the conditioning videos used for reproducibility, the `SceneLayout` JSON, the `Perturbation` delta, the post-generation HS axes, the seed-provenance trace (which seed, which models, which checkpoints, which git commit), and *format adapters* that repackage the bundle in nuScenes, Waymo, and OpenSCENARIO formats. Customers consume in their native format; the adapter layer makes the synthetic data drop-in compatible with existing mmdetection3d / OpenPCDet training pipelines.

TABLE I
VALIDATION RECIPES — ALL ON BEVFUSION (LiDAR + CAMERA).

Recipe	Training data
A	nuScenes train (real-only baseline).
B	A + 10K <i>random</i> Cosmos generations from random scenes.
C	A + 10K Cosmos generations balancing nuScenes class distribution.
D	A + 10K of <i>verified high-signal</i> synthetic from WOD-E2E seeds (this work).
E	A + 10K from same source seeds as D but skipping Stage D filter (proves filtering matters).

IX. VALIDATION PROTOCOL

A. The Flagship Experiment

The empirical contribution rests on a controlled ablation that compares four data designs on the same baseline detector and the same held-out long-tail eval (Table I). Pretraining is BEVFusion-from-scratch with the standard nuScenes recipe; each design is retrained for 20 epochs on 8×H100s, three seeds each.

Pass criteria (registered in advance):

- **D vs A:** $\geq +5\%$ relative NDS on the long-tail eval. Demonstrates that synthetic data helps at all.
- **D vs B:** $\geq +3\%$ relative NDS. Demonstrates that targeting matters.
- **D vs C:** $\geq +2\%$ relative NDS. Demonstrates that long-tail focus, not just class balance, matters.
- **D vs E:** $\geq +2\%$ relative NDS. Demonstrates that the filter contributes signal beyond Cosmos generation.

If all four pass, the program has product-market-fit evidence. If only D vs A passes, the program has a paper but a weaker thesis about which contribution is load-bearing.

B. Secondary Validation: Closed-Loop on Bench2Drive

A secondary experiment trains UniAD-Tiny / VAD on real-only versus real + HS-synthetic and runs both on the 220 Bench2Drive routes [25]. Reports Driving Score, Success Rate, and per-scenario breakdown (cut-ins, unprotected turns, occluded VRUs). Pass criterion: real + synthetic $\geq$ real-only by $\geq 5\%$ Driving Score, especially on long-tail scenarios.

C. Quality Audits

Parallel to the headline experiments: (i) 200 random verified clips human-rated on a 1–5 realism scale by 3 raters, target mean ≥ 3.8 ; (ii) Cosmos-Reason2-8B Gate B match rate logged across the run, target $\geq 85\%$; (iii) UniAD-derived auto-labels manually checked on 100 clips, target 3D-box recall $\geq 90\%$; (iv) cross-sensor consistency — project camera detections into synthetic LiDAR, target $\text{IoU} \geq 75\%$.

TABLE II
MODEL STACK. EVERY MODEL IS PRETRAINED AND OPEN-WEIGHT.

Role	Checkpoint
Embed (visual)	DINOv3 ViT-L/16 [11]
Embed (video)	V-JEPA-2.1 ViT-G/16 [12]
Detector ensemble	UniAD-Base, VAD-Base, BEVFormer-Base [24]
Tagger / Verifier	Cosmos-Reason2-8B [10]
Generator (single-view)	Cosmos-Transfer-2.5-2B [9]
Generator (8-cam)	Cosmos-Transfer1-7B-Single2MultiView
Generator (LiDAR)	Cosmos-Transfer-LidarGen
Auto-labeler	UniAD pipeline (3D + tracks + occupancy)
Eval baseline	BEVFusion (LiDAR + camera)

TABLE III
BULK PROCESSING — TOTAL H100-HOURS FOR THE FULL V0 PASS ON THE WOD-E2E CORPUS.

Phase	H100-hr
Mining all of Tier-0 (~1.4M clips, one-time)	140
Layout extraction (Tier-0)	2,000
Generation: WOD-E2E (4,021 seeds × 20 variants)	9,400
Filtering all generated outputs	3,000
Validation (BEVFusion × 5 recipes × 3 seeds)	10,000
Subtotal — V0 (WOD-E2E only)	~24,500

X. DATASETS, MODELS, AND COMPUTE

A. Seed Sources (Stage A inputs)

Tier 0 (must-have): nuScenes [21], Waymo Open + WOD-E2E [22], CODA [23], comma2k19, NHTSA SGO crash CSVs (~2.6 TB local). Tier 1 (V0.5): Argoverse 2, IDD-3D, DAIR-V2X, DENSE/CADC, nuPlan (+1.8 TB).

B. Models — All Pretrained

C. Compute Footprint

Per-seed steady-state on 1×H100 with the 2.5-2B primary: Stage A ~0.4 sec, Stage B ~15 sec, Stage C ~2–3 hr for 20 variants (single + multi-view + LiDAR + auto-label), Stage D ~5 min. Roughly ~3 hr per seed for 20 variants → ~10 verified clips per H100-hour at the assumed 50% gate-survival rate.

Storage: ~10 GB per verified clip uncompressed × 40K ≈ 400 TB, or ~80 TB with Draco LiDAR compression. With shared institutional GPU access, the program is feasible without external compute funding; with no shared cluster, the validation phase (Phase P5) is the only step that requires planning a dedicated allocation.

XI. PHASED PLAN

The full program is structured as a 12-week core build plus 6 weeks of ensembling and customer-loop hooks (Table IV). Phases are independent: a partial deployment (Phases 0–4) already produces the headline ~40K verified clips and the

TABLE IV
PHASED PLAN, SCOPED TO A FULL RESEARCH PROGRAM.

Wk	Phase / Milestone
1	P0: cluster + storage provisioned; Cosmos smoke-test on sample data.
2–3	P1: Stage A — embed, ensemble, tag, score Tier-0; PostgreSQL seed table populated; oracle smoke-test passes.
4–5	P2: Stage B — SceneLayout extractors for Waymo, CODA, nuScenes, nuPlan, comma2k19 at scale.
6–7	P3-1: Perturbation generator + Cosmos pipeline driver + auto-labeler; end-to-end test on 50 seeds × 20 variants.
8	P3-2: Inspect 100 random generations; tune perturbation ranges.
9	P4: Stage D filter live on 100 seeds; expect ~500 verified outputs.
10	P5-1: Headline downstream training experiment (Table I, 5 recipes).
11	P5-2: Scale to all 4,021 WOD-E2E seeds; expect ~40K verified outputs; HF dataset release.
12	P5-3: Format adapters (nuScenes, Waymo, OpenSCE-NARIO); end-to-end provenance audit.
13–14	P6-1: Add MagicDrive-V2 and Vista as ensemble alternatives.
15–16	P6-2: Customer-model failure gate via on-prem container; closed-loop measurement protocol.
17–18	P6-3: 2–3 design-partner pilots; paper submission; documentation.

validation table; Phases 5–6 add ensembling, customer-loop, and design-partner pilots.

XII. RISKS AND MITIGATIONS

Cosmos Transfer doesn’t preserve trigger event timing precisely (e.g., pedestrian dart shifted by ±2s vs. intended). Mitigation: encode trigger timing in the cuboid sequence with sub-frame precision; filter at Gate B; if widespread, fall back to Cosmos-Predict-2 with stronger temporal prior [9].

Generated LiDAR quality is poor enough that downstream LiDAR detectors don’t generalize. Mitigation: ship synthetic LiDAR with a quality flag; for camera-only customers this is moot; for LiDAR-heavy customers, fall back to camera-only synthetic plus the customer’s own LiDAR sim.

Sim-to-real gap. Mitigation: use mixed real-synthetic training (10–30% synthetic only); aggressive filtering; downstream eval gates inclusion.

Filter survival rate is much lower than 50%. Mitigation: tune perturbation aggressiveness; default to smaller deltas first, escalate if survival is high; per-seed-class monitoring.

Compute footprint overshoots. Mitigation: Cosmos-Transfer-2.5-2B is 3.5× smaller than 7B and outperforms it on AV [9] — already the primary path. Cache aggressively. Use shared institutional GPU access for non-validation phases.

Provenance bug ships data labeled with the wrong seed. Mitigation: comprehensive integration tests; nightly canary that traces 100 random outputs back to seed and verifies

correctness; full git-commit + checkpoint-version stamp on every bundle.

Seed coverage gaps for ultra-rare scenarios (flooded road, fire) absent from any seed dataset. Mitigation: use Cosmos-Reason2-8B to generate text-only seed descriptions; condition Cosmos-Predict-2.5 (text-to-video) for these; lower confidence flag on the output.

XIII. CONCLUSION

Long-tail VLM driving performance is the right problem to be working on in 2026, and the right approach is to close the loop between mining, generation, and verification rather than to scale any one of those alone. The proposed program builds, validates, and releases that loop end-to-end at production scale, with a 5-axis calibrated scorer, multi-view + LiDAR generation, four filter gates, and a registered ablation table that quantifies how much each component contributes. The output — ~40K verified high-signal synthetic clips with provenance — would, on its own, more than double the open-source long-tail AV training-data supply. The closed loop is the moat: targeting plus filtering, neither of which the existing literature provides.

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